GREGORY C. CHANG, PhD, MPH

Health Economics & Outcomes Research (HEOR) | Real-World Evidence (RWE) | Epidemiology

Columbus, OH | 614-736-1251 | gcchang03@gmail.com | LinkedIn | Google Scholar | gregory-chang-epi.github.io

PROFESSIONAL SUMMARY

HEOR/RWE scientist with a PhD in Epidemiology and 7+ years of observational research across medical devices, pharmaceuticals, and public health. Lead end-to-end real-world evidence studies on administrative claims data (Komodo Health) for cardiac devices, from protocol and SAP through analysis and deliverables for market access, payer, medical affairs, and executive audiences. First or co-author on 11 peer-reviewed publications.

CORE COMPETENCIES

Study Design	Retrospective cohort; treatment patterns; burden of illness; HCRU and cost; comparative effectiveness and safety; protocol and SAP development
Methods	Survival analysis; propensity score / IPTW; confounding and bias assessment; complex survey analysis; pharmacovigilance
Data & Tools	Claims data (Komodo Health; ICD-10/CPT/HCPCS); R, SAS, SQL/Snowflake, Power BI, Python; REDCap; AI/LLM-assisted evidence synthesis
Therapeutic Areas	Cardiac devices and heart failure; oncology; immunology; infectious disease

PROFESSIONAL EXPERIENCE

ZOLL Cardiac Management Solutions, Columbus, OH **Apr. 2026 – Present**

Health Economics Specialist

- Lead **real-world evidence studies** on the Komodo Health claims database for ZOLL's cardiac device portfolio—**LifeVest** wearable cardioverter defibrillator, **Heart Failure Management System (HFMS)**, and **TherOx SuperSaturated Oxygen (SSO2) Therapy**—evaluating utilization, clinical outcomes, and healthcare resource use.
- Author **study protocols and statistical analysis plans**; define claims-based cohorts, exposures, and outcomes (ICD-10, CPT, HCPCS); apply survival and comparative methods to produce evidence for **market access, payer, and medical affairs** audiences.
- Conduct **payer authorization and market analyses** that quantify barriers to patient access and inform leadership's payer advocacy and competitive strategy.
- Design and maintain **Power BI dashboards** that convert internal operational and commercial data into decision-ready business intelligence for executives and cross-functional teams.
- Develop an internal **AI/LLM evidence-repository tool** over peer-reviewed literature to accelerate evidence retrieval and synthesis across the organization.

Analysis Group, New York, NY **Jun. 2025 – Aug. 2025**

Associate Intern, Health Economics and Outcomes Research

- Led **retrospective cohort analyses** in relapsed/refractory **multiple myeloma** evaluating real-world effectiveness and safety of newly approved bispecific antibodies—adverse events, time to discontinuation, progression-free and overall survival—and delivered client-ready R Markdown reports.
- Integrated **clinical trial and Komodo Health real-world data** to characterize treatment patterns and healthcare utilization in support of **market access and regulatory** strategy for oncology and immunology assets.
- Designed **feasibility assessments** and data-collection instruments for **chart review studies**, reducing reliance on external vendors.
- Conducted an **AI/LLM-assisted targeted literature review** of inflammatory bowel disease treatments to inform pharmacoepidemiologic and market access strategy.

The Ohio State University Comprehensive Cancer Center – James, Columbus, OH **Aug. 2023 – Dec. 2024**

Susan G. Komen TREND Trainee, Center for Cancer Health Equity

- Led analyses of **breast cancer screening adherence** before, during, and after the COVID-19 pandemic using national BRFSS data; presented findings at AACR 2024.

Nationwide Children's Hospital, Center for Injury Research and Policy, Columbus, OH **May 2022 – Jan. 2024**
Project Manager, NIH R01

- Managed an **NIH R01** trial of mobile technology interventions to reduce distracted driving; supervised a **6-member multidisciplinary team**, ran data operations in REDCap, and co-authored two peer-reviewed publications.

U.S. Centers for Disease Control and Prevention, Atlanta, GA **Aug. 2018 – Jan. 2022**
Epidemiologist (PHI/CDC Global Health Fellow), Division of Global HIV & TB

- Supported national **Population-based HIV Impact Assessments (PHIA)** in Kenya, Nigeria, Lesotho, Uganda, and Zimbabwe; analyzed **complex survey data** from 25,000+ participants per survey to validate indicators for government-published final reports.
- Led analyses on viral load suppression, hazardous alcohol use, and HIV awareness across 6–13 countries, resulting in **four first- or co-authored publications** in JAIDS, JIAS, and BMC Public Health.
- Collaborated with Ministries of Health and implementing partners, including a prior year of field data operations for CAMPHIA 2017 with ICAP at Columbia University (Yaoundé, Cameroon).

U.S. Centers for Disease Control and Prevention, COVID-19 Response, Atlanta, GA **Jun. 2020 – Mar. 2021**
Analyst, SARS-CoV-2 Reinfection and Vaccine Breakthrough Unit

- Managed the national clinician-reported **SARS-CoV-2 reinfection investigation**, building the REDCap case-reporting system and specimen-handling SOPs; co-authored findings in Clinical Infectious Diseases.
- Served as data manager for **vaccine breakthrough surveillance**, contributing to the MMWR report of the first 10,000+ breakthrough cases.

EDUCATION

The Ohio State University — PhD, Epidemiology (GPA 3.90) May 2026
Dissertation: Structural and spatial factors associated with HIV viral load suppression in East Africa: evidence from population-based surveys

Columbia University Mailman School of Public Health — MPH, Epidemiology 2018

Case Western Reserve University — BA, Biology & Anthropology 2016

HONORS

Dean's Award for Public Health Impact, The Ohio State University College of Public Health 2026
Susan G. Komen Training to Eliminate Disparities (TREND) Fellowship, OSUCCC – James 2023 – 2024
Public Health Institute / CDC Global Health Fellowship 2018 – 2022

PEER-REVIEWED PUBLICATIONS

1. **Chang GC**, Bingham T, et al. (2025). Undetectable HIV-1 viral load among virally suppressed people living with HIV: implications for Undetectable equals Untransmittable (U=U) in resource-limited settings using dried spot testing. *JAIDS*. DOI: 10.1097/QAI.0000000000003745
2. **Chang GC**, Ramesh B, Liebhard B, Borrasca WE, Loux T, Horan J, Turner AN. (2024). Census tract-level social vulnerability and gonorrhea rates in central Ohio: results from the STD Surveillance Network, 2020–2021. *Sexually Transmitted Diseases*. DOI: 10.1097/OLQ.0000000000002062
3. Zhang F, Benedetti M, **Chang G**, Evans A, Pan Y, Shoots-Reinhard B, Zhu M. (2024). Demographic characteristics and trends of cell phone use while driving citations in selected states in the United States, 2010–2020. *Traffic Injury Prevention*, 1–7. DOI: 10.1080/15389588.2024.2351605
4. Evans A, Benedetti MH, **Chang G**, et al. (2024). A comparison of licensed driver counts from the Federal Highway Administration and driver licensing agencies in 11 states. *Traffic Injury Prevention*, 25(2):133–137. DOI: 10.1080/15389588.2023.2278412
5. Ewing AP, **Chang GC**, et al. (2023). Lifestyle related cancer risk and protective behaviors vary among a convenient sample of physically active, young-to-middle-aged adults 18–49. *Int J Environ Res Public Health*, 20(13):6305. DOI: 10.3390/ijerph20136305
6. **Chang GC**, West CA, et al. (2022). Hazardous alcohol use and HIV indicators in six African countries: results from the Population-based HIV Impact Assessments, 2015–2017. *J Int AIDS Soc*, 25(11):e26029. DOI: 10.1002/jia2.26029

7. Achia T, Cervantes IF, Stupp P, Musingila P, Muthusi J, Waruru A, Schmitz M, Bronson M, **Chang G**, et al. (2022). Methods for conducting trends analysis: roadmap for comparing outcomes from three national HIV population-based household surveys in Kenya (2007, 2012, and 2018). *BMC Public Health*, 22(1):1337. DOI: 10.1186/s12889-022-13633-8
8. West CA, **Chang GC**, et al. (2021). Unawareness of HIV infection among men aged 15–59 years in 13 sub-Saharan African countries: findings from the Population-Based HIV Impact Assessments, 2015–2019. *JAIDS*, 87(Suppl 1):S97–S106. DOI: 10.1097/QAI.0000000000002708
9. CDC COVID-19 Vaccine Breakthrough Case Investigations Team (contributing author). (2021). COVID-19 vaccine breakthrough infections reported to CDC — United States, January 1–April 30, 2021. *MMWR Morb Mortal Wkly Rep*. DOI: 10.15585/mmwr.mm7021e3
10. Lee JT, Hesse EM, Paulin HN, Datta D, Katz LS, Talwar A, **Chang G**, et al. (2021). Clinical and laboratory findings in patients with potential SARS-CoV-2 reinfection, May–July 2020. *Clin Infect Dis*, 73(12):2217–2225. DOI: 10.1093/cid/ciab148
11. Zheng L, Jones C, Ejigu G, George N, Geller A, **Chang GC**, et al. (2021). Countries with delayed COVID-19 introduction – characteristics, drivers, gaps, and opportunities. *Global Health*, 17:28. DOI: 10.1186/s12992-021-00678-4

CONFERENCE PRESENTATIONS

1. **Chang GC**, Fisher JL, Paskett ED. Disparities in mammography screening adherence throughout phases of the COVID-19 pandemic: results from the 2018, 2020, and 2022 Behavioral Risk Factor Surveillance System. 17th AACR Conference on the Science of Cancer Health Disparities, Los Angeles, CA, September 2024.
2. **Chang GC**, Ramesh B, Liebhard B, Turner AN. Census tract-level social vulnerability and gonorrhea in Ohio: results from the STD Surveillance Network 2020–2021. International Society for Sexually Transmitted Diseases Research, Chicago, IL, July 2023.
3. **Chang GC**, O'Connell C, et al. The association between alcoholic substance abuse and key HIV outcomes: results from six population-based HIV impact assessments in sub-Saharan Africa. 14th INTEREST Conference on HIV Treatment, Pathogenesis, and Prevention, Windhoek, Namibia. Oral abstract, November 2020.
4. **Chang GC**, Bingham T, et al. Can Undetectable equals Untransmittable (U=U) be applied in resource-limited settings: results of the Population-based HIV Impact Assessment surveys in eleven African countries, 2015–2018. Poster (1082), Conference on Retroviruses and Opportunistic Infections (CROI), Boston, MA, March 2020. Abstract

TECHNICAL REPORTS

1. Ohio Commission on Minority Health. (2025). *Medical Expert Panel on Preventable Cancers: White Paper – Achieving equity and eliminating preventable cancer disparities within racial and ethnic populations* (Vol. 3). Contributing analyst. Report
2. **Zimbabwe Population-based HIV Impact Assessment 2020 Final Report.** Technical assistance for field training and data analysis. Report
3. **Kenya Population-based HIV Impact Assessment (KENPHIA) 2018 Final Report.** Technical assistance for survey implementation and data analysis. Report
4. **Nigeria HIV/AIDS Indicator and Impact Survey (NAIIS) 2018 Final Report.** GIS-based field deployment planning. Report

LEADERSHIP & PROFESSIONAL SERVICE

Board Member, Survivorship Committee — James Next-Generation Ambassadors, OSUCCC – James 2022 – Present
Peer Reviewer — Advances in Drug and Alcohol Research; Sexually Transmitted Diseases; BMC Infectious Diseases; BMC Public Health